Bihar Engineering University, Patna

B.Tech 3rd Semester Examination, 2025

Course: B.Tech
Code: 105304

Subject: Discrete Mathematics and Graph Theory

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option / answer the following (Any seven question only)*** **[2 x 7 = 14]**

(a) If A={1,2,3}, how many subsets does A have?
(i) 3
(ii) 8
(iii) 6
(iv) 9

(b) Principle of induction is used to prove statements about
(i) Real numbers
(ii) Complex numbers
(iii) Natural numbers
(iv) Integers only

(c) The base step in induction verifies
(i) All cases
(ii) Recursive step
(iii) First case
(iv) Final step

(d) $p \rightarrow q$ is false when
(i) p = T, q = T
(ii) p = T, q = F
(iii) p = F, q = T
(iv) p = F, q = F

(e) Contrapositive of $p \rightarrow q$ is
(i) $\neg p \rightarrow \neg q$
(ii) $q \rightarrow p$
(iii) $\neg q \rightarrow \neg p$
(iv) $p \rightarrow \neg q$

(f) A group must satisfy
(i) Closure only
(ii) Closure + Associativity
(iii) Closure + Associativity + Identity + Inverse
(iv) Commutativity

(g) Homomorphism preserves
(i) Elements
(ii) Size
(iii) Structure
(iv) Order only

(h) A graph with no cycles is called
(i) Complete graph
(ii) Bipartite graph
(iii) Tree
(iv) Directed graph

(i) Maximum edges in a simple graph with n vertices
(i) n
(ii) n^2
(iii) $n(n-1)/2$
(iv) 2n

(j) Degree sum formula states
(i) Sum of degrees = n
(ii) Sum of degrees = edges
(iii) Sum of degrees = 2 × edges
(iv) Sum of degrees = vertices²

Q.2 (a) Prove by induction: $\frac{1}{1.2} + \frac{1}{2.3} + \cdots \frac{1}{n(n+1)} = \frac{n}{(n+1)}$ **[7]**

(b) Solve the recurrence relation $y_{n+2} - 5y_{n+1} + 6y_n = 5^n$ subject to the condition $y_0 = 0, y_1 = 2$. **[7]**

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Q.3 (a) Prove that every finite subset of a lattice has an LUB and a GLB. [7]

(b) In a Lattice if $a \leq b \leq c$, then show that $(a \vee b) \vee (b \wedge c) = (a \vee b) \wedge (a \vee c) = b$. [7]

Q.4 (a) State and prove Lagrange's theorem for group. Is the converse true? [7]

(b) Let G be the set of all non-zero real number and let $a * b = ab/2$. Show that $(G,*)$ be an abelian group. [7]

Q.5 (a) Prove that a simple graph with n vertices and k components can have at most $\frac{(n-k)(n-k+1)}{2}$ edges. [7]

(b) Prove that a connected graph G is Euler graph if and only if every vertex G is of even degree. [7]

Q.6 (a) Let $(A, \leq)$ be a partially ordered set. Let $\leq$ be a binary relation on A such that for a and b in A, a is rerated to b iff $b \leq a$. Show that $\leq$ is partially ordered relation. [7]

(b) Show that $\Big((p \vee q) \wedge \sim (\sim p \wedge (\sim q \vee \sim r))\Big) \vee (\sim p \wedge q) \vee (\sim p \vee r)$. Is a tautology without using the truth table. [7]

Q.7 (a) If $f: A \to B$, $g: B \to C$ are invertible functions, then show that $gof: A \to C$ is invertible and $(gof)^{-1}: f^{-1}og^{-1}$. [7]

(b) Determine whether the following function is a bijection from R to R: $f(x) = \frac{(x2+1)}{(x2+2)}$. [7]

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Q.8 (a) Rewrite the following arguments using quantifiers, variables and predicate symbols: [7]

(i) All birds can fly

(ii) Some men are genius

(iii) Some numbers are not rational

(b) Write the symbolic form and negate the following statements: [7]

(i) Everyone who is healthy can do all kinds of work

(ii) Some people are not admired by everyone

(iii) Everyone should help his neighbours, or his neighbours will not help him

Q.9 Write short note on ***any two*** of the following: [7 x 2 = 14]

(a) Euler Graph and Hamiltonian Graph

(b) Homomorphism and Isomorphism Graph

(c) Planar and Complete Bipartite Graph

(d) Adjacency matrix of a graph

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